

Project Management Case Study

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I. Project Charter

Project Title: Mobile and Web-based Event Layout Planner with 3D Visualization and Preview.

Project Manager: Ms. De Jesus, Sherina Ann M.

Objectives

1. To design and develop a mobile and web application with a 2D/3D editor, preview functionality, ready-made venue and theme templates, and object placement tools.
2. To implement a system that converts 2D layouts into optimized 3D models.
3. To enable users to share and publish layouts online for collaboration and export visual access.
4. To integrate a cost estimation feature that calculates total event expenses based on asset pricing.
5. To implement an event scheduling system where clients reserve a time slot in affiliated venues. Once a schedule exists on a date, the layout for that venue is locked from edits, ensuring no double-booking or conflicts.
6. To provide users an editor that allows to customize their own layout.
7. To enable the system to automatically generate and adjust venue layouts and capacity estimates according to the number of guests provided.
8. To integrate an AI-assisted layout design feature that proposes and designs table and chair arrangements with optimized capacity, sightlines, and flow.

Scope (Inclusions & Exclusions)

Inclusions

- User access for event organizers and affiliated venues.
- 2D/3D layout editor with object placement tools (chairs, tables, stage, booths, etc.).
- Ready-made theme and venue templates.
- Conversion of 2D layouts into optimized 3D previews.
- Cost estimation for assets (e.g., Wooden Chair – ₱500, Metal Chair – ₱1,000).
- Event scheduling with affiliated venues where organizers reserve time slots.
- AI-powered tool that helps design and suggest table and chair layouts.

Exclusions

- Advanced CAD-level precision or support for all professional file formats.
- AR overlays using live camera feeds (limited only to 3D preview, not AR).
- Integration with external CAD tools, acoustics, or lighting simulation.
- Real-time collaborative editing across multiple devices.
- Structural safety checks or compliance with specific building codes.
- Offline functionality.

Stakeholders

Owner of the Event Company

Responsible for overseeing the overall business operations and ensuring that the system contributes to improved efficiency, reliability, and service quality within the company.

Event Organizer / Event Planner (Staff)

Utilizes the system to design, edit, validate, and manage event layouts, as well as to reserve schedules with affiliated venues to prevent conflicts and ensure proper coordination.

Affiliated Venue (Owner / Representative)

Manages the availability of the venue, confirms reservations made by event organizers, and ensures that the approved layouts comply with venue policies and standards.

Event Company Customer (Client)

Engages with the system by requesting event bookings, reviewing proposed layouts, and providing feedback or approval for final event setups.

Constraints

1. The system must be developed and deployed within the academic/project timeframe.
2. The system must be fully functional by Term 1 of the academic year 2026-2027.
3. The project operates under a limited budget for software tools, 3D assets, and cloud hosting services, constrained by the student developers' financial resources.
4. The platform must be mobile and web accessibility.
5. The system's performance may depend on device specifications, particularly when rendering 3D layouts.
6. Venue scheduling is restricted to affiliated venues only.
7. An internet or Wi-Fi connection is required to access the application.

Methodology/Approach

Scrum Agile Methodology was chosen due to its flexibility and ability to gain feedback through its iterative nature. This allows greater transparency between developers and users along with being able to enhance communication between everyone involved with the project. It also reduces risks like large scale failures and allows for continuous improvements due to retrospection from each sprint.

II. Work Breakdown Structure (WBS)

1.0 Initiation

- 1.1 Problem, Objectives, and Scope Identification
- 1.2 Stakeholder Identification
- 1.3 Feasibility Study
- 1.4 Project Proposal Approval

2.0 Planning

- 2.1 Conduct Stakeholder Interview
- 2.2 Requirements Gathering
- 2.3 Technical Design & Architecture
- 2.4 System Design & Prototyping
- 2.5 Project Schedule
- 2.6 Risk Assessment and Management Plan
- 2.7 Budget Plan
- 2.8 System Analysis and Design Approval

3.0 Execution

- 3.1 Final System Design
- 3.2 Backend & Database setup
- 3.3 Frontend Development
- 3.4 Mobile Development
- 3.5 Web Development

3.6 AI Integration

3.7 System Integration

3.8 Devops & Security

3.9 Internal Testing and Debugging

3.10 Documentation of the Development Process

4.0 Testing & Quality Assurance

4.1 Test Plan & Device Matrix (QA & Beta)

4.2 Sprint Testing & Bug Tracking

4.3 Closed Beta & Feedback Collection

4.4 Bug Fixing and Issue Iterations

4.5 ISO/IEC 25010 Quality Evaluation

4.6 User Acceptance Testing

4.7 Documentation of the Testing Process

5.0 Closure

5.1 Final Review

5.2 Final Documentation

5.3 Final Presentation

5.4 System Deployment, Documentation, and Project Asset Handover

5.5 Retrospective & Archive

III. Project Schedule (Gantt Chart & PERT Analysis)

3.1. Gantt Chart:

Task	Start Date	End Date	Duration	Dependency
Project Initiation	August 4, 2025	August 10, 2025	6 Days	
Data and Requirements Gathering	August 12, 2025	August 24, 2025	12 Days	Project Initiation
Requirement Analysis	August 25, 2025	August 27, 2025	2 Days	Data and Requirements Gathering
Project Proposal Approval	September 5, 2025	October 9, 2025	4 Days	Requirement Analysis
Workflow and Data Diagrams	September 11, 2025	September 20, 2025	9 Days	Project Proposal Approval
Stakeholder Approval	September 25, 2025	November 6, 2025	20 Days	Workflow and Data Diagrams
System Design				
Architecture	November 8, 2025	November 15, 2025	6 Days	Stakeholder Approval
Database	November 16, 2025	November 20, 2025	4 Days	Architecture
Scalability Planning	November 21, 2025	November 25, 2025	4 Days	Database
System Design Presentation	November 26, 2025	December 2, 2025	8 Days	Scalability Planning
System Design Revisions	December 3, 2025	December 20, 2025	17 Days	System Design Presentation
Development Phase				
System Development	January 5, 2026	March 15, 2026	50 Days	System Design Revisions
Backend Implementation	March 17, 2026	April 15, 2026	28 Days	System Development
System Development Presentation	April 16, 2026	April 20, 2026	4 Days	Backend Implementation
Development Revisions	April 21, 2026	May 6, 2026	15 Days	System Development Presentation
Testing Phase	May 8, 2026	June 15, 2026	7 Days	Development Revisions
Deployment Phase	June 18, 2026	September 22, 2026	30 Days	Testing Phase
Post-Launch				
System Monitoring	June 18, 2026	September 22, 2026	30 Days	Deployment Phase

Documentation & Closure	September 15, 2026	September 22, 2026	7 Days	System Monitoring
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3.2. PERT Analysis:

Activity	Predecessor (s)	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected Time (TE)	Critical Path?
A - Project Initiation		1	3	5	3.0	Yes
B - Data and Requirements Gathering	A	1	3	5	3.0	Yes
C - Requirement Analysis	B	2	6	7	5.5	Yes
D - Project Proposal Approval	C	1	3	5	3.0	Yes
E - Workflow and Data Diagrams	D	2	3	6	3.3	Yes
F - Stakeholder Approval	E	1	3	5	3.0	Yes
System Design						
G - Architecture	F	2	5	7	4.83	Yes
H - Database	G	2	5	7	4.83	Yes
I - Scalability Planning	H	2	5	7	4.83	Yes
J - System Design Presentation	I	1	3	5	3.0	Yes
K - System Design Revisions	J	2	4	6	4	Yes
Development Phase						
L - System Development	K	5	10	15	10.0	Yes
M - Backend Implementation	L	5	9	10	8.5	Yes
N - System Development Presentation	M	1	3	5	3.0	Yes
O - Development Revisions	N	5	8	9	7.6	Yes
P - Testing Phase	O	2	3	5	3.16	Yes
Q - Deployment Phase	P	3	3	5	3.33	Yes
Post-Launch						
R – System Monitoring	Q	1	3	5	3.0	Yes

S - Documentation & Closure	R	1	3	5	3.0	Yes
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IV. Risk Management Plan

Risk	Probability	Impact	Mitigation
Server Failure during Operations	Medium	High	Regular Backups and Use of Local Servers.
Bugs on Launch	Low	High	Final Testing and Contingencies.
Performance Issues with Low End Devices	High	High	Optimize 3D Rendering with LoD (Level of Detail), caching and fallback 2D view for low-end devices.
WebGL Performance/Load Issues	High	High	Usage of Unity WebGL Compression, providing a Lite 2D Mode fallback for low-end browsers and Lazy-loading 3D Assets.
Build Size Too Large	High	Medium	Strip unused Unity Modules (Physics and etc), Compressing Texture assets, Use of Addressables for asset streaming.
Cross Platform UX Problems	Medium	Medium	Separation of control schemes for mobile and web along with responsive UI design in Unity UI Toolkit/Canvas.
Reverse Engineering of WebGL Builds	Medium	Medium	Obfuscate WebGL builds with Tools and Keep sensitive logic server-side.

V. Resource & Communication Plan

Team Members and Roles:

Project Manager: De Jesus, Sherina Ann M.

Project Leader: Emmanuel Josef P. Siva

Full Stack Developer: Alejo, John Benedict S.

Quality Assurance (QA) Tester: De Jesus, Sherina Ann M.

Backend Developer: Punzalan, Maverick P.

Database Management: Siva, Emmanuel Josef P.

Documentation:

Punzalan, Maverick P.

Siva, Emmanuel Josef P.

Communication Tools & Frequency:

Weekly Meetings: MS Teams / Google Meet / Discord

- These platforms will be used for virtual team discussions, progress updates, issue resolution, and planning for upcoming tasks.

Instant Communication: Facebook Messenger

- Messenger will serve as a quick-response platform for time-sensitive questions or announcements outside formal meetings.

Formal Updates: Email

- Emails will be used to send progress reports, formal announcements, and communications to external stakeholders or advisors.

Task & Project Management: Notion

- Tasks, deadlines, and progress will be tracked through boards and timelines in Notion.

File Storage & Sharing: Drive / OneDrive

- These cloud platforms will store documents, images, presentations, and reports accessible to all team members.

VI. Budget & Cost Estimation

Project Budget Estimation:

Item (s)	Estimated Cost (₱)
Web Hosting & Server (Staging + CDN)	₱ 5,000.00
AI Integration	₱ 3,000.00
Developer Stipend (Team)	₱ 18,000.00
Tools / Services (CI, Real Time Trial, Licenses)	₱ 2,000.00
Testing & QA	₱ 3,000.00
Contingency	₱ 5,000.00

Total Estimated Cost: ₱ 31,000.00

The estimated budget for the Mobile and Web-based Event Layout Planner with 3D Visualization and Preview is subject to change based on project requirements, system enhancements, and actual implementation costs. Expenses may vary depending on hosting providers, AI service usage, and additional features integrated during development.

Cost and Benefit Analysis:

The following table outlines the projected benefits of implementing the Mobile and Web based Event Layout Planner with 3D Visualization and Preview.

Item (s)	Type	Estimated Value (₱)	Notes
Printing / Papers Savings.	Tangible	₱ 5,000 per year	Less paper for onsite plans and brochures.
Manual Labor Savings.	Tangible	₱ 3,000 per event	Fewer manual labor adjustments and fewer staff hours.
Reusable Templates and Asset.	Tangible	₱ 5,000 over 3 years	Reused across upcoming events.
Increased Participation / Convenience	Intangible		Easier access may raise adoption rates.
Improved Transparency and Trust	Intangible		Audit logs and previews increase stakeholder confidence.
Enterprise Sales / Upsell Potential	Tangible / Intangible		Enables venue/chain contracts.

Summary:

- **Total Tangible Benefits in Year 1:** ₱ 8,000.00
- **Total Tangible Benefits in 3 Years:** More than ₱ 25,000.00
- **Project Cost:** ₱ 31,000.00
- **Break-even Point:** After 4 years, the project starts generating profit or value beyond its total expenses.

VII. Monitoring & Change Control

The following tools and log entries illustrate how monitoring and change control are applied to the EventScape Project.

Monitoring Tools Used:

Gantt Chart

The Gantt Chart was used to visually track project tasks, durations, and deadlines. It allowed the team to compare planned versus actual progress, monitor overlapping activities, and manage resource allocation. This ensured that milestones were met on time and deliverables followed the project timeline.

PERT Chart

The PERT Chart complemented the Gantt Chart by focusing on task dependencies and critical paths. It helped identify potential delays, prioritize essential activities, and adjust task sequences to minimize risks and ensure continuous progress.

Weekly Updates & Meetings

Team members share progress, report issues, discuss goals, and approve adjustments to keep the project on track.

Unity Profiler

The Unity Profiler monitored CPU/GPU usage, memory consumption, draw calls, and rendering performance. It helped identify issues such as lag, overheating, and battery drain, allowing developers to optimize performance across devices.

Firebase Crashlytics

The Firebase Crashlytics collected real time crash reports, stack traces and error logs from mobile builds. It highlighted device specific failures and frequent issues, enabling the team to prioritize bug fixes and improve system stability.

Server Monitoring

The Server Monitoring tracked uptime, response times, API performance, and load balancing. It ensured smooth synchronization when offline users reconnected and maintained system reliability with minimal downtime.

Bug Tracker

The Bug Tracker was utilized to document, categorize, and monitor software defects throughout development. It allowed the team to assign issues to responsible members, set priorities, and track resolution progress.

Testing Logs

Documents all executed test cases, expected versus actual results, and any defects or issues identified during quality assurance. It also tracks severity, status, and steps taken to reproduce issues, providing a comprehensive record to help developers prioritize fixes and ensure system reliability.

Change Control Process:

Since the project is still in its early stages and has not yet entered the full development phase, the following entries are provided as examples of what a change log might contain. These are assumptions and serve only as illustrations of how modifications, improvements, and fixes would be documented during later stages of the project.

Sample Change Log Entries

Change Request	Date	Status	Decision
Replace 3D Library for performance improvements	September 14, 2025	Under Review	Pending technical feasibility analysis
Update color scheme for better accessibility	September 20, 2025	Rejected	Does not align with current design guidelines
Enable multi-user real time editing	September 25, 2025	Approved	Scheduled for v2.0 roadmap

VIII. Project Closure Report & Final Presentation

Since the project is still in its early stages, this section will contain anticipated deliverables and performance expectations.

- **Planned Deliverables**
 - System Design for both Mobile and Web-based Project.
 - Client Collaboration on the Project.
 - Working Prototype for Mobile and Web-based Applications.
 - Proposed Engine for 2D/3D Viewing.
 - Comprehensive project documentation.
- **Projected Performance**
 - **Scope:** Complete Delivery of Planned and Additional Features.
 - **Time:** Completion before the end of the first semester of Academic Year 2026–2027, prior to the conclusion of Capstone 2.
 - **Cost:** A budget of ₱ 31,000.00
- **Anticipated Risks or Challenges**
 - Time constraints and potential development delays caused by overlapping academic requirements and limited project periods.
 - Finding a suitable engine for the 3D view.
 - System bugs and performance issues during testing and deployment.
 - Technical difficulties during implementation.
- **Initial Lessons learned (if any)**
 - Prepare a backup plan for certain deliverables if not met within the timeframe.
 - Conduct further data gathering to fully assess the project's feasibility.
 - Ensure consistent collaboration and communication among team members and stakeholders to ensure successful project outcomes.
- **To be Updated**
 - Potential adjustments to the project's scope, including additions or removals of features, based on further data gathering, feasibility evaluations, and feedback from panelists.